

Electric Air  
Shipping

# JELLYFISH

A MONTHLY UPDATE

NO. 002 - SEPTEMBER 2026



# EAS is decarbonising global freight with ultra-heavy-lift cargo airships.

## IN THIS ISSUE

|                  |                                  |                     |                                 |                                         |                                           |
|------------------|----------------------------------|---------------------|---------------------------------|-----------------------------------------|-------------------------------------------|
| 01               | 02                               | Shape, and<br>VS/VE | 03                              | 04                                      | 05                                        |
| Asked to present | The design goes<br>international |                     | Modern<br>Airships: a<br>review | End of an era:<br>Leaving the<br>office | Missed the Boat:<br>Freight<br>Nightmares |

## 01 - ASKED TO PRESENT

### 48 hours - two engineering societies and the old school

Things seem to happen in waves sometimes. During September, within 48 hours, Toby and Brian were asked to present the EAS project to two separate engineering societies, and Toby's old high school.

The first is the **Royal Aeronautical Society**, Adelaide Chapter, on 8 October. The RAeS is a prestigious global society, whose London chapter is currently working with the Airship Association on future partnerships. The enthusiasm from the RAeS was humbling and we can't wait to highlight the key aspects of the project.

The second is the **Sustainable Engineers Society** (SENG) in November, to also be streamed to the Australian and international chapters. We have been told that 'engineers love equations' and again can't wait to demonstrate some of the unique aspects of the project to a technical audience.

#### INDUSTRY INSIGHT



#### Decarbonising Global Freight with Cargo Airships

##### PRESENTERS

**Brian and Toby Brandenburg** (Founders)  
Electric Air Shipping (EAS)



 **Thursday 8 October, 2026**  
5:30 pm - 7:30 pm

 **UniSA**, Building MM1-05,  
University Boulevard,  
Mawson Lakes, SA 5095



And on 19 October, Toby will be presenting at his old high school – St Michael's College, to their year 10 Aerodynamics and Environmental Chemistry stream – focusing on curiosity, self-belief and big ideas as we approach a decarbonised future.

## 02 - THE DESIGN GOES INTERNATIONAL

### 电动航空货运私人有限公司

During September, EAS formally applied for the Sunship's registered design to be extended to China, USA and EU, meaning the goal for a strong international IP portfolio is well underway.

A late flurry as key paperwork was lodged in the final days before priority expiration proved a learning experience and a critical task achieved. The Chinese character EAS name is above.

As the Sunship's design is critical in its function as a technology, there is a great overlap of IP protection, a very exciting position for our (currently) humble startup.

#### THE IP STACK

**Registered Design** – The Sunship

**Patent** (pending) – The Sunship

– VS/VE based airship modelling software

**TM** – The Sunship

*With more to come*

# Shape is shape, regardless of size

*Technically, an airship could be any shape...*

Because no shape holds more volume than a sphere (for a given prime dimension), and aerostatic lift is proportional to enclosed volume...

We say: a useful cargo airship's shape should be close to a sphere, while maintaining some 'inherent directionality'. (And aerodynamics can be improved separately with other systems)

'VS/VE' is our simple framework for measuring potential airship envelope shapes. Because it is ratio based, it describes the shape, regardless of size. Here's the general idea:

**VS** = the product (multiplication) of the shape's 3 main aspect ratios. A measure of isotropy...

$VS_{\infty}$  = the spherical average of VS from all of the shape's aspect ratios.

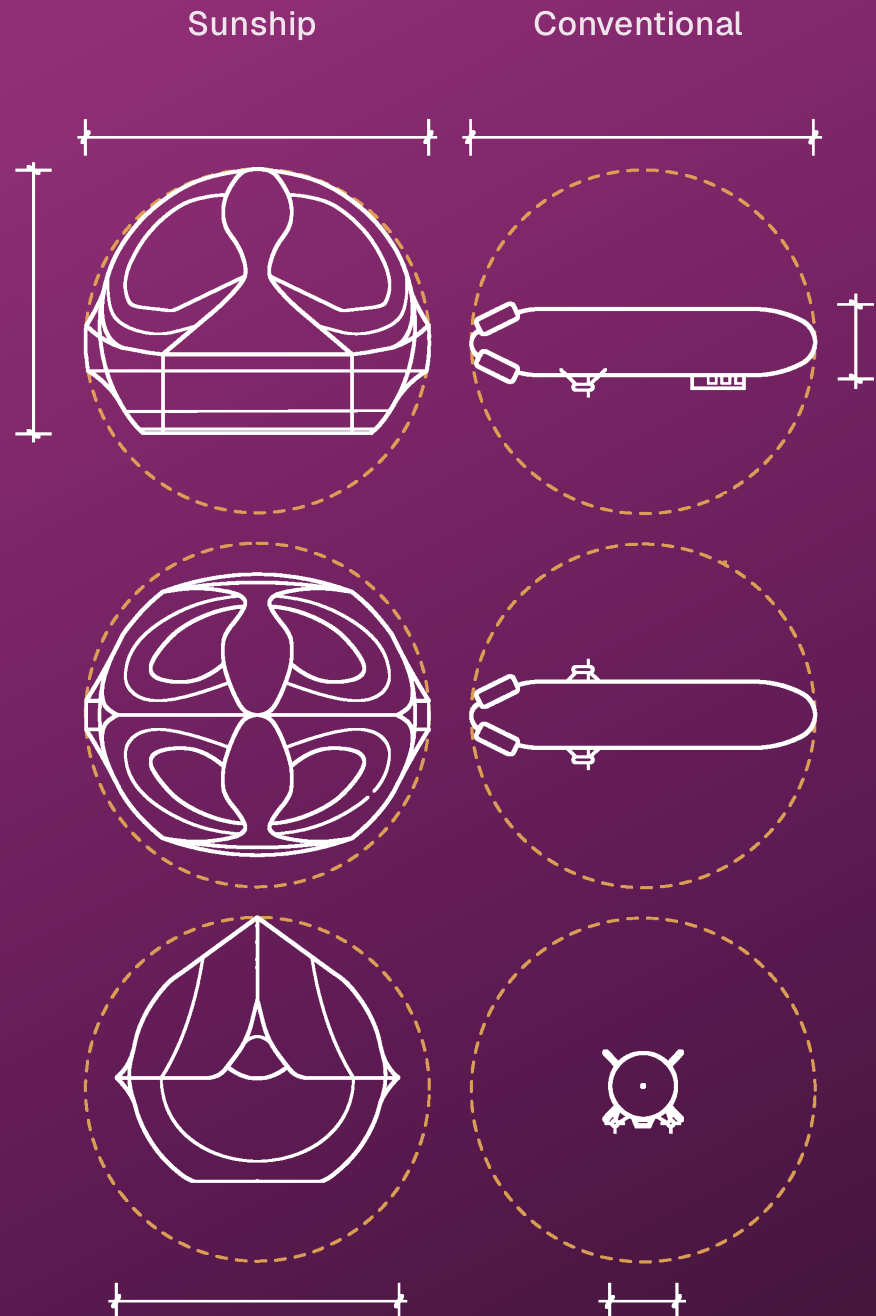
**VE** = what % of a sphere does this shape fill?

VS and VE are generally inversely correlated, but not perfectly so. Only a perfect sphere has a VS = 1 and a VE = 100%.

If we choose the correct shape - at giant industrial size ranges, the differences are significant.

*Everything has a shape...*

*So, VS/VE imagines every single thing in the universe, spinning inside of a ball.*



EAS airship form

$1.28 \times 1.21 \times 1.05$

**VS 1.63**

**VE 53%**

Cigar airship

$5 \times 5 \times 1$

**VS 25**

**VE 3%**

'Ratio Airship' shows the Sunship is ~20x more volume efficient than a conventional cigar airship at any size - despite the square cube law acting equally upon both.

## 03 – THE PRIOR ART

## MODERN AIRSHIPS, PARTS 1-3

## Modern Airships is an extraordinary catalogue of more than 285 historic and modern airship concepts

Published by the Lyncean Group, collated by Peter Lobner. After two centuries of experimentation, it is very difficult to find anything in airship design that has not been tried before: almost every imaginable configuration appears somewhere in the record... Almost every.

Perhaps most surprising is how counter-intuitive the shape solution is. Conventional airship thinking naturally prioritises streamlined shapes, high-lift helium and low drag; the Sunship starts instead with volume efficiency and useful ultra-heavy lift, then solves the resulting aerodynamic problems around them.

The field accumulated most of the necessary ingredients, but seemingly never reorganised the design problem around volume efficiency. VS/VE exposes that missing optimisation axis; the Gömböc provides a remarkable geometric answer; and the Sunship is the systems architecture that attempts to make that answer operational.

CHATGPT, WHEN ASKED TO COMPARE MODERN AIRSHIPS WITH THE EAS KNOWLEDGE BASE

You can access Modern Airships here: [lynceans.org/all-posts/modern-airships-part-1](https://lynceans.org/all-posts/modern-airships-part-1)

## 04 – END OF AN ERA

## Moving out of the original EAS office

Brian and Toby moved into the original Frome Street EAS office in 2022. During this time the space has become a comfortable and creative haven, and a magical space, housing the formulation of VS/VE, the

Sunship conceptual design, and the birth of the EAS project in its current form.

Sadly the lease is ending, and EAS is moving onwards to new things – next to be housed in Stone & Chalk Adelaide.



---

## 05 – MISSED THE BOAT

- **The pump sprayers** – a company selling consumer garden sprayers had been awarded lucrative shelf space at a massive Australian retailer. The popularity of the product meant at times the shelves were near empty, forcing quick air freight to maintain supply. A garden sprayer is *literally* empty volume – and so perhaps the ultimate item to be punished by air freight's 'chargeable weight' penalty – to the point where air freight shipping costs were literally equal to the cost of the product. A situation where the shipper was desperate for a roomy middle shipping option without volumetric penalties.
- **The transformer internals** – fridge-sized, heavy, dense electronic components. Sea freight suited the majority of the time, and long lead times not a problem. Except for when sudden demand spikes hit – causing the factory to work triple shifts, and units to be air-freighted at massive cost. The company CEO's own words – 'direct to LA or Thailand would be amazing', dreaming of adaptable routing, a key feature of the high-cadence middle option.
- **The Pallet Line** – If a pallet of goods sells for less than X (around \$1000), air freight is simply not viable. Not only this, but manufacturing decisions are made accordingly. A Sunship-based middle option not only adds flexibility and reduces costs for stakeholders, but also allows greater flexibility for manufacturing and warehousing decisions.

If you have had shipping horror stories, let us know! [contact@electricairshipping.com](mailto:contact@electricairshipping.com)

---

## BRIAN'S THOUGHTS

# serendipity

/sɛrˈɒn-dɪpˈɪ-tē/

*“An unplanned fortunate discovery.”*

We are pleased to present Jellyfish number 002, covering the start to the end of September 2026.

Since 2002, this invention and design project has had a string of instances of serendipity.

An early instance was my realisation that the Sunships would not compete with passenger aircraft, and that instead they could compete with huge, heavy surface ships – potentially driving them off the seas and into the history books, with the assistance of the International Maritime Organization, nations and world governments. That was such a leap of scale: from a minor invention to a whole new industry. I look forward to writing more about this topic in future, and addressing appropriate gatherings of interested people.

Serendipity has continued in September, with the enthusiastic organisers of two unrelated associations of engineers contacting us to invite us to speak at their monthly meetings in the near future – in October and early November.

Also very fortunately, as the four-year lease on our much-loved initial company office comes to an end in October this year, a family member has been able to secure a multi-roomed suite of offices right here in the city for a reasonable rent. That will enable us to base ourselves at Stone & Chalk, and also use a room at the new office in central Adelaide for free if needed – all at a time when budgeting and fundraising have become top priorities.

More serendipity has come from a meeting Toby had a few days ago

with Stone & Chalk, to help us cope with a difficult transition stage. The AI-generated summary has been very helpful, to my mind, in setting priorities for October in particular, and then moving on to November and the end of the year.

The time since May, and moving out of years of research and design, has been very circular and full of novelty and uncertainty. In that time, like so many projects, the long term seemed inspirational and a massive benefit to humanity, the middle term was full of amazing designs and business opportunities, and the short term was just plain worrying.

That can change now.

## TOBY'S THOUGHTS

It might not seem like a massive news item, but moving out of the office, for me as the human, is probably the biggest story of the month.

It's not exactly a surprise, because the terms of the lease were always coming up to around about this time. But there's that feeling in life when doors open and close, and chapters open and close. And moving out means reflecting on moving in, which was four years ago now, but in a sense sort of feels like yesterday.

So maybe let's just back it up to when we moved in.

### Take the sign down right now and I'll take this office.

It was 2022. EAS as an entity didn't exist, although as a concept it had existed for a long time. The office was very, very daggy and dusty, and a terrible off-cream colour. I remember painting all of the walls myself. And, look, this is Toby's Thoughts, and the painting reminds me of the emotions of it all, so let me talk about it from the personal perspective.

The sweet story is that at that time I had a girlfriend who I loved dearly, and she was interested in painting lessons. There was a lovely lady next door named Janet, who very sadly passed away recently, and that really adds to the sense of closure around this chapter. She was very, very sweet, and she was instrumental in us getting this office – and, in that sense, instrumental in the whole EAS project, because this office itself became instrumental in the research breakthroughs, just by giving us a space to focus on the work. I spoke to Janet about painting lessons for my girlfriend at the time, and she said what she always said to anyone who enquired about lessons: "Sorry, there's no room, but I think one lady is going to be away next week, so you can squeeze in then. We're actually full up right now, I'm very sorry."



**And of course, that was a total lie. There was always room,** but they really liked their little group of painting friends, and so they were quite protective of who they let in. Literally as I left after talking to Janet that day, I saw the property manager placing the For Lease sign in the window of the door, and I just said to him, "Take the sign down right now and I'll take this office."

And that's sort of how it happened.

So, you know, you do things with good intentions, and sometimes unexpected consequences come as a result. I think I'm a big believer in that – that putting things out there can lead to mysterious, unexpected coincidences. That was 2022, and then sadly, not too long afterwards, that girlfriend and I parted ways and I was quite heartbroken. Still, the office remained.

## TOBY'S THOUGHTS

CONTINUED

It converted me from a sceptic into a believer.



I recall vividly painting the walls of the office here – a beautiful pinky-cream colour on one half and bright orange on the other – tears in my eyes, some sad music playing, up on the ladder by myself late into the night. I remember bringing all of the furniture in and slowly setting the place up over the next few years.

When Brian and I moved in, I was still focused mostly on my previous life and career as a teacher and music educator. I was running private music tuition out of the space. I was also running a whole bunch of computers – a little server farm – getting into AI at the very early stages of image diffusion, and generally running a thousand little schemes hoping something would work out, while working in a school and juggling some other casual work. I had this feeling that by around 2023 or 2024, when the first two-year lease expired, that *surely* one of these schemes would work out, at least enough that I would be able to consolidate and focus.

Little did I know that, along the way, this crazy, ludicrous airships idea that Brian and I were working on would actually produce the breakthroughs that it did: the Gömböc, VS/VE, and everything that went along with them. It converted me from a sceptic into a believer. EAS went from being a passion project, a hobby and an interest for both myself and Brian, to the point where, around 2024, I left essentially every other part of my working life behind to focus on it beyond full-time. **So it's been an amazing four-year run.**

Moving out is a little bit sad, but now that we're members of Stone & Chalk, and have access to some other spaces as well, the office has become a little redundant. And it just feels like about the right time. Four years. That's like a university degree. That's a stage of life.

So I guess I'll wrap up this section by reiterating a few things.

Firstly, a goodbye to Janet, the painter from next door, an incredible artist and just a lovely lady. I used to really enjoy my conversations with her and her husband John next door, who runs a bookshop. Just lovely people.

And secondly, I suppose, the point that sometimes in life, if you approach things from a good place, really good things can come from it in unexpected ways. I distinctly remember telling Janet and others, as we moved into this office, that I was going to pursue this crazy airships idea of my dad's as hard as I could. I was really going to try my best to make something come out of it. And I distinctly remember saying that I didn't really have any fear of failure, because ultimately it was his idea, not mine. Something about having that little bit of separation from the idea gave me an enormous freedom around it – boundless creativity, really, and very little fear of failure.



## TOBY'S THOUGHTS

CONTINUED

## We are humans behind this project, and at this point it is still very much a father-son project.

So if you've made it this far, thanks for listening to my thoughts. We are humans behind this project, and at this point it is still very much a father-son project. We're very early in what is essentially a gigantic, probably decades-long undertaking, but I don't know what else I'd rather be doing, to be honest. There are good things ahead, and it feels like a time of change for us. At the end of this four-year stage, we're leaving stealth mode, leaving the office,

and going out into the world. That's happening figuratively, but it's also happening quite literally.

So maybe the final thought is for you, dear reader, to think about place and space, **your connection with where you are, and the intentions that brought you to the place you're in.** And drop me a line if any of this resonates with you.

And, look, we'll be back next month to do it all again.

PEACE - TOBY B



Electric Air Shipping Pty Ltd  
SE 2 L 1, 157 Frome St  
Adelaide SA 5000  
Kaurua Land

Jellyfish No.002  
September 2026  
Published 1 October 2026

electricairshipping.com  
Written and edited by Toby and Brian  
Brandenburg. Assembled with Claude. Nothing  
goes out that neither of us has read end to end.

FEED ME TO YOUR AI → [electricairshipping.com/llms-full.txt](https://electricairshipping.com/llms-full.txt)